

4.4 Written Descriptions of a Sequence of Transformations

Lesson Introduction

 Benchmarks	MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates. MA.912.GR.2.5 Given a geometric figure and a sequence of transformations, draw the transformed figure on a coordinate plane.		 Learning Targets	- I can draw an image given a preimage and a written description of a sequence of transformations on a coordinate plane. - I can describe a transformation on the coordinate plane using written descriptions.
 Benchmark Coherence	Building On MA.8.GR.2.3 Describe and apply the effect of a single transformation on two-dimensional figures using coordinates and the coordinate plane.		Working Toward N/A	
 Essential Questions	- How can I use the information from a written description of a series of transformations to draw an image? - How can I apply transformations to a table of values to help me draw an image?			
 Lesson Narrative	In the lesson, students will continue their work with sequences of transformations but will be transitioning into drawing the image itself rather than just identifying transformations. Students will be given a sequence of transformations and learn to draw the new image that results after a sequence of transformations is applied to the preimage. Students evaluate tables of values in order to explore a variety of representations in transformations.			
 Component Summary	4.4.1 Warm-Up (~5 min.)	Notice and Wonder about the pattern from a kaleidoscope (<i>SE p. 196</i>)	4.4.3 Guided Instruction (15 min.)	Use coordinates to draw an image as the result of a given sequence of transformations (<i>SE p. 198</i>)
	4.4.2 Collaboration (15 min.)	Draw the transformation of images on a coordinate grid (<i>SE p. 196</i>)	4.4.4 Your Turn! (10 min.)	Use coordinates to draw an image as the result of a given sequence of transformations (<i>SE p. 200</i>)
	4.4.5 Wrap-Up (~5 min.)	Draw an image on the coordinate plane that is the result of a given sequence of transformations (<i>SE p. 201</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Sequence of transformations <u>Additional Terms:</u> - Reflection - Rotation - Translation		(Optional) Patty paper (Optional) Graphing utility (Optional) Colored pencils	
				Additional Preparation N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may swap the x- and y-values with the coordinates when graphing. - Students may reverse the signs when completing rotations. - Students may misunderstand the direction of clockwise and counterclockwise.	Precision matters in this lesson as students begin drawing on a coordinate plane. Rotations and reflections may be hard for students to “see” and even harder to produce on their own. Visit the 4.5.1 Warm-Up for Lesson 5 as a gauge for what students might be able to produce and which elements of precision to highlight and spiral throughout this lesson.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
 Differentiate Instruction	Notice and Wonder The 4.4.1 Warm-Up provides students with a colorful pattern from a kaleidoscope to determine things that they notice and things that they wonder. Consider using responses related to symmetry, reflections, or similar geometric terms as meaningful connections throughout this lesson and unit.	MA.K12.MTR.4.1: Discussion and Reflection 4.4.2 Collaboration provides students with an opportunity to engage in collaborative discussion and reflection with their partner about drawing an image on the coordinate grid given a sequence of transformations.
	For a kinesthetic entry point for all transformations in this lesson, provide copies or patty paper for students to manipulate based on the given transformation. This will allow students to demonstrate their understanding of the transformation prior to drawing it on the grid paper, where mistakes may happen. For example, utilize Desmos or GeoGebra to illustrate each rotation, translation, or reflection in real-time while combining this with student-created foldables or anchor charts for each.	
Lesson Notes:		

4.4.1 Warm-Up: Notice and Wonder

Component Overview: Students will complete a Notice and Wonder activity about the pattern from a kaleidoscope.



4.4.1 Warm-Up: Notice and Wonder

A kaleidoscope is an instrument that contains loose bits of colored material, usually glass, between two flat plates and two plane mirrors. The plates can be moved so the colored material changes position. The mirrors then reflect the material in an endless variety of patterns.



This Photo by Unknown Author is licensed under CC BY

1. What do you notice about the pattern from the kaleidoscope in the picture?
Answers will vary. The image has symmetry across any line drawn from the tip of one yellow point through the center to the tip of the yellow point across from it.
2. What do you wonder about the pattern?
Answers will vary. Is there symmetry across any lines drawn through the center?



Optional questions to consider if time permits:

- *What geometric shapes do you notice in this image?*
- *How do you think the kaleidoscope creates these shapes without having those shapes on the device?*
- *How do you think it is possible for the colors to appear the way that they do in this image?*
- *Does this make you think any differently about how you view transformations like reflections?*

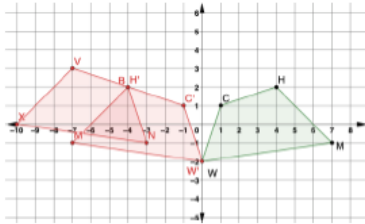
4.4.2 Collaboration: Drawing the Image

Component Overview: Students will work together to draw images on a coordinate grid using their knowledge of transformations. Students should work individually to first draw the transformations before checking their work with their partner.



4.4.2 Collaboration: Drawing the Image

1. Quadrilateral $CHMW$ is shown on the coordinate grid.



- a. Draw the quadrilateral that is the result of a reflection across the y -axis and then translated left 3 and up 1. Label the new quadrilateral $BVXN$.

Shown on graph.

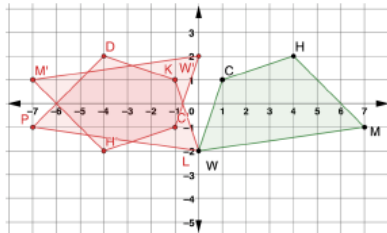
- b. Check your graph with your partner. Make any necessary adjustments.

A **sequence of transformations**, or a composition of transformations, is when two or more transformations are combined.

- c. Work with your partner to explain why quadrilateral $BVXN$ is a result of a sequence of transformations.

Answers will vary. The quadrilateral is first reflected and then translated, so there is more than one transformation to get the new quadrilateral $BVXN$.

2. Quadrilateral $CHMW$ is shown on the coordinate grid.



This component is designed to lead students through a discovery process. Students may benefit from using colored pencils to show the figure after each transformation.



Consider prompting struggling learners to map one vertex at a time or use patty paper.



Consider the use of colored pencils and highlighters if using a document camera or other display.

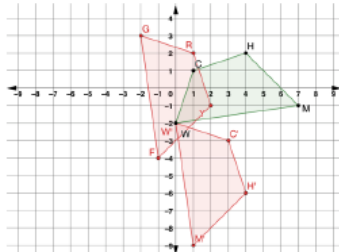
4.4.2 Collaboration: Drawing the Image (continued)

- a. Draw the quadrilateral that is the result of a rotation of 180° counterclockwise about the origin and then reflected across the x -axis. Label the new quadrilateral $KDPL$.

Shown on graph.

- b. Check your graph with your partner. Make any necessary adjustments.

3. Quadrilateral $CHMW$ is shown on the coordinate grid.



- a. Draw the quadrilateral that is the result of a rotation of 90° clockwise about the point W and then translated left 2 and up 5. Label the new quadrilateral $RYFG$.

Shown on graph.

- b. Check your graph with your partner. Make any necessary adjustments.

4. For each sequence of transformations, summarize your mistakes and what you learned from them.

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Start
To
Acquire
Knowledge
Experience
Skills

- a. A reflection across the y -axis and then translated left 3 and up 1
b. A rotation of 180° counterclockwise about the origin and then reflected across the x -axis
c. A rotation of 90° clockwise about the point W and then translated left 2 and up 5
Answers will vary.



Monitor for students or partners with different images and have them compare and contrast their images.



For an extension or to challenge students who finish early, ask students to explain and represent how a figure that is not rotated about the origin could be translated to the origin, rotated, then translated again to “undo” the first translation.

A sample response: Translate the figure by translating the center of rotation, Point W , up two units to be located on the origin. Then follow the algebraic descriptions for rotations listed in Lesson 1. Then translate the rotated figure back down two units.



Allow students to share out their mistakes in a whole-group setting. When student(s) share their mistake(s), poll the class to determine who shares in that mistake(s). This can help students realize that they are not alone in common errors, so they feel encouraged to keep trying. If no student wants to share, list common mistakes you noticed and wrote down during your monitoring of the students' work during the activity. Strategies like this and reflection opportunities like mistakes cultivate a classroom culture of growth mindset and perseverance. Even consider a prize or celebration for students who share the most common mistake.

4.4.3 Guided Instruction: Using Coordinates

Component Overview: Students will receive guided instruction on the use of coordinates to draw an image as the result of a given sequence of transformations. For this activity, consider using the gradual release model of “I Do” and “We Do” throughout.



4.4.3 Guided Instruction: Using Coordinates

The Effect of Transformations on Coordinates

Rotations Clockwise About the Origin		
Rotations of 90°	Rotations of 180°	Rotations of 270°
Change the x -coordinate to its opposite, then switch the coordinates	Change both coordinates to their opposite	Change the y -coordinate to its opposite, then switch the coordinates
Rotations Counterclockwise About the Origin		
Rotations of 90°	Rotations of 180°	Rotations of 270°
Change the y -coordinate to its opposite, then switch the coordinates	Change both coordinates to their opposite	Change the x -coordinate to its opposite, then switch the coordinates

Translations	
Right	Left
Add to the x -coordinate	Subtract from the x -coordinate
Up	Down
Add to the y -coordinate	Subtract from the y -coordinate

Reflections	
x -axis	y -axis
The y -coordinate changes to its opposite.	The x -coordinate changes to its <i>opposite</i> .
$y = x$	$y = -x$
The coordinates switch.	The coordinates switch and become their opposite.



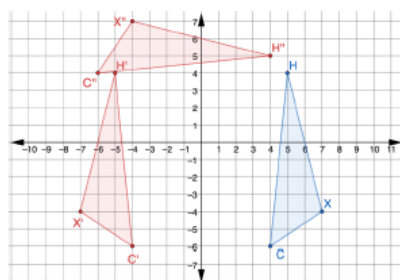
Instruction begins with a written description of the algebraic description of different transformations. To help students connect the algebraic description, have the students write the algebraic description under each written description.



Consider creating this page as a note-catcher sheet or an anchor chart that students can reference or utilize in later lessons throughout the unit, or have students create their own anchor charts personalized to their entry points and learning needs.

4.4.3 Guided Instruction: Using Coordinates (continued)

1. Triangle CHX is shown on the coordinate grid.



Triangle CHX

Vertex	x	y
C	4	-6
H	5	4
X	7	-4

- a. Reflect $\triangle CHX$ across the y -axis then rotate 90° clockwise about the origin. Show both $\triangle C'H'X'$ and $\triangle C''H''X''$.

Shown on graph.

- b. Complete the table of values for each vertex.

Triangle $C'H'X'$

Vertex	x	y
C'	-4	-6
H'	-5	4
X'	-7	-4

Triangle $C''H''X''$

Vertex	x	y
C''	-6	4
H''	4	5
X''	-4	7

- c. Discuss with your partner how the tables show the effect of transformations on coordinates.

Answers will vary. The numbers in the table matched the directions from the guidelines for the given transformation.



- How is the written description of the effect of a reflection across the y -axis seen in the table of values?
- How is the written description of the effect of a rotation of 90° clockwise about the origin seen in the table of values?



For a visual entry point, have students first graph the values from the table. Then encourage students to anticipate where each point will lie (which quadrant) after each translation.

4.4.3 Guided Instruction: Using Coordinates (continued)

One strategy to apply transformations is to use tables of ordered pairs and then to plot the ordered pairs.

2. The table of values shows the coordinates of the vertices of quadrilateral $WBKT$.

Quadrilateral $WBKT$

Vertex	x	y
W	-17	-5
B	-11	-5
K	-4	-11
T	-14	-16

- a. Complete the table of values for the given sequence of transformations of quadrilateral $WBKT$.
Rotate quadrilateral $WBKT$ 180° counterclockwise about the origin, then translate it right 7 down 4.

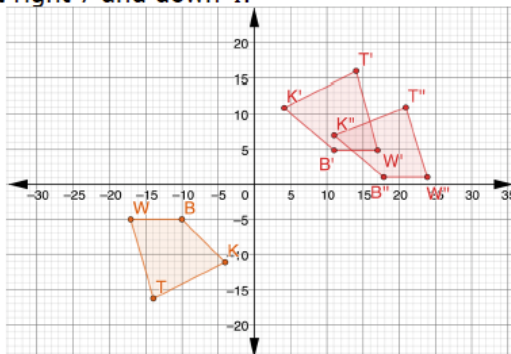
Quadrilateral $W'B'K'T'$

Vertex	x	y
W'	17	5
B'	11	5
K'	4	11
T'	14	16

Quadrilateral $W''B''K''T''$

Vertex	x	y
W''	24	1
B''	18	1
K''	11	7
T''	21	12

- b. Quadrilateral $WBKT$ is shown on the coordinate grid. Show the sequence of transformations on the coordinate grid by rotating quadrilateral $WBKT$ 180° counterclockwise about the origin, then translating it right 7 and down 4.



For students who struggle with tables of values, completing the tables before graphing will be harder. For students who understand transformations and algebraic descriptions, using tables is usually easier to do first before graphing. Question 2 allows students with these strengths to understand that their method is just as effective as graphing. Allow students' choice in which entry point works best for them as they navigate this activity.



Consider using physical manipulatives to assist students with visualizing what this sequence of transformations would look like. For a different visual, consider using a graphing technology such as Desmos or GeoGebra to support their drawings.

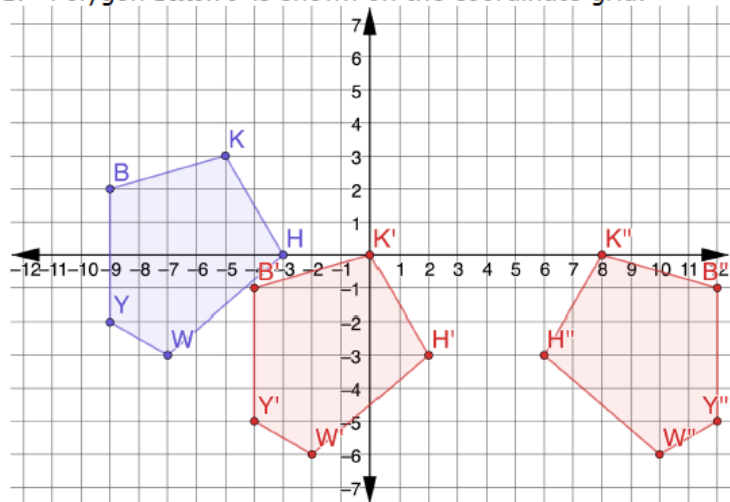
4.4.4 Your Turn!

Component Overview: Students will engage in an independent practice on the use of coordinates to draw an image as the result of a given sequence of transformations. This activity should be done independently to allow students to practice.



4.4.4 Your Turn!

1. Polygon $BKHWY$ is shown on the coordinate grid.



On the coordinate grid, translate polygon $BKHWY$ right 5 and down 3. Then, reflect the result across the line $x = 4$. Label the resulting polygon $B''K''H''W''Y''$.

Shown on graph.



Consider using the time for this activity to work with small groups of students who may need additional support based on their performances during this lesson. Small groups should consist of students who may struggle with transformations at a single step. With these students, review each transformation one at a time, ensuring they understand each part before moving on. During this time, you may determine which students struggle with specific transformations and provide more targeted support.



When reviewing this activity, consider asking students if they used a table of values, mapped the transformations, or use patty paper to help determine the location of the image.

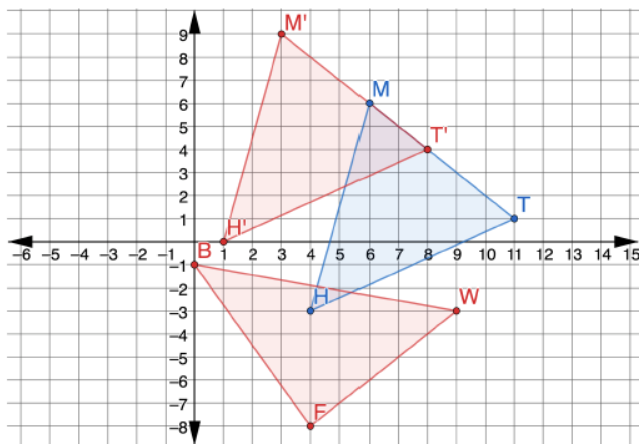
4.4.5 Wrap-Up: Ticket Out the Door

Component Overview: Students will complete a Ticket Out the Door that assesses student understanding of drawing an image on the coordinate plane that is the result of a given sequence of transformations. This activity should be done independently to gather quantitative data on whether or not students understand the content of this lesson.



4.4.5 Wrap-Up: Ticket Out the Door

- Triangle HMT is shown on the coordinate grid.



Draw $\triangle BWFF$, which is the result of translating $\triangle HMT$ left 3 and up 3, then rotating the result 90° clockwise about the origin.

Shown on graph.



For additional insight, consider having students use two different colors to perform each transformation separately. Although this is one sequence of transformations, students drawing each step will provide additional information for future remediation.